

AI Response Evaluation Study

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AI Response Evaluation Study

1. Executive Summary

This project evaluates the quality of AI-generated responses using a structured human evaluation framework. The study is designed to assess how effectively AI models respond to a diverse set of user prompts across areas including general knowledge, science, history, technology, business, education, writing, reasoning, instruction following, and ambiguous questions.

A total of **50 prompts** are evaluated, with **two AI-generated responses per prompt**, resulting in **100 responses** for evaluation. Each response is assessed using six criteria: accuracy, relevance, completeness, clarity, instruction following, and overall quality.

Each criterion is scored on a **1–5 scale**, where 1 represents very poor performance and 5 represents excellent performance. The evaluation results are used to identify strengths, weaknesses, recurring errors, and differences in response quality between the evaluated AI models.

The purpose of this study is not to establish a universal ranking of AI models, but to demonstrate a structured and reproducible approach to human evaluation of AI-generated content.

2. Project Objective

The primary objective of this project is to develop practical experience in evaluating AI-generated responses using consistent quality standards.

The specific objectives are:

1. Evaluate AI responses for factual accuracy and reliability.
 2. Determine whether responses directly address the user's request.
 3. Assess whether responses provide sufficient information to satisfy the prompt.
 4. Evaluate clarity, organization, and readability.
 5. Determine whether AI models follow explicit instructions provided in prompts.
 6. Compare the performance of two AI models using the same evaluation framework.
 7. Identify recurring response-quality problems.
 8. Develop a structured dataset containing evaluation scores and qualitative comments.
 9. Practice consistent human judgment and evidence-based evaluation.
 10. Produce a documented evaluation report suitable for inclusion in an AI evaluation portfolio.
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3. Dataset

3.1 Dataset Overview

The evaluation dataset consists of **50 user prompts** covering 10 different categories.

Category	Number of Prompts
General Knowledge	5
Science	5
History	5
Technology	5
Business	5
Education	5
Writing	5
Reasoning	5
Instruction Following	5
Ambiguous Questions	5
Total	50

Each prompt is submitted to two AI models using the same wording.

Therefore:

50 prompts × 2 AI responses = 100 responses

3.2 Dataset Fields

The evaluation dataset contains the following fields:

- Prompt ID

- Category
- Prompt
- AI Model
- AI Response
- Accuracy Score
- Relevance Score
- Completeness Score
- Clarity Score
- Instruction-Following Score
- Overall Quality Score
- Evaluator Comments

3.3 Dataset Design

The prompts were designed to test different aspects of AI performance rather than focusing on a single type of question.

The dataset includes factual questions, explanatory questions, writing tasks, reasoning problems, instruction-based tasks, and questions where additional context may be required.

This variety helps provide a broader assessment of response quality.

4. Evaluation Methodology

The evaluation follows a standardized human-evaluation process.

Step 1: Prompt Creation

Fifty prompts were created across ten categories.

Step 2: Response Generation

Each prompt was submitted independently to two AI models using identical wording.

Step 3: Response Collection

The complete responses generated by both models were recorded in a structured spreadsheet.

Step 4: Independent Evaluation

Each response was evaluated using a predefined rubric.

The evaluator assigned a score from 1 to 5 for each evaluation criterion.

Step 5: Qualitative Analysis

In addition to numerical scores, comments were recorded to explain important strengths, weaknesses, errors, or instruction-following issues.

Step 6: Model-Level Analysis

The scores were aggregated by model to calculate average performance for each evaluation criterion.

Step 7: Error Analysis

Responses were reviewed to identify recurring issues such as factual errors, incomplete answers, irrelevant information, poor clarity, and failure to follow instructions.

Scoring System

Each criterion uses the following scale:

Score	Meaning
1	Very Poor
2	Poor
3	Acceptable
4	Good
5	Excellent

The evaluation is based on predefined criteria rather than personal preference.

5. Evaluation Criteria

5.1 Accuracy

Accuracy measures whether the information provided by the AI is factually correct.

Score 1: Major factual errors or misleading information.

Score 2: Multiple factual problems.

Score 3: Mostly accurate but contains minor issues.

Score 4: Accurate with only minor imperfections.

Score 5: Fully accurate based on available evidence.

5.2 Relevance

Relevance measures whether the response directly addresses the user's request.

A highly relevant response stays focused on the question and avoids unnecessary information.

Score 1: Does not address the question.

Score 2: Mostly unrelated.

Score 3: Partially addresses the question.

Score 4: Directly addresses the question with minor unnecessary content.

Score 5: Direct, focused, and highly relevant.

5.3 Completeness

Completeness measures whether the response adequately addresses all important aspects of the prompt.

Score 1: Almost entirely incomplete.

Score 2: Major information is missing.

Score 3: Partially complete.

Score 4: Mostly complete.

Score 5: Fully addresses the requirements of the prompt.

5.4 Clarity

Clarity measures how easily a reader can understand the response.

Factors considered include:

- Organization
- Grammar
- Readability

- Logical structure
- Appropriate language
- Explanation quality

Score 1: Extremely difficult to understand.

Score 2: Frequently unclear.

Score 3: Understandable but has noticeable clarity problems.

Score 4: Clear and well organized.

Score 5: Extremely clear, concise, and easy to understand.

5.5 Instruction Following

Instruction following measures whether the AI correctly follows the specific requirements contained in the prompt.

For example, if a prompt requests:

"Explain the topic in exactly 100 words using five bullet points."

The evaluator checks whether the response actually follows those requirements.

Score 1: Completely ignores instructions.

Score 2: Follows very few instructions.

Score 3: Follows some instructions.

Score 4: Follows most instructions.

Score 5: Fully follows the stated instructions.

5.6 Overall Quality

Overall quality represents the evaluator's assessment of how useful and effective the response is as a complete answer.

The overall score should consider the other evaluation criteria rather than being based solely on personal preference.

6. Results

This section should be completed only after the evaluation of all 100 responses.

6.1 Overall Results

Evaluation Criterion	Model A	Model B
Accuracy	4.5	4.5
Relevance	4.6	4.5
Completeness	4.8	4.9
Clarity	4.5	4.4
Instruction Following	4.9	4.9
Overall	4.6	4.6

Scores represent the average rating on a 1–5 scale.

6.2 Category-Level Results

Category	Model A	Model B
General Knowledge	5	5
Science	5	5
History	5	[Score]
Technology	5	5
Business	5	4.5
Education	4	5
Writing	5	5
Reasoning	5	5
Instruction Following	5	4

7. Model Comparison

The two AI models are compared using the same prompts and evaluation criteria.

The comparison should focus on observed performance within this specific dataset rather than making broad claims about either model.

Accuracy

Model A: Model A's response does contain misinformation about GK questions but it provides an accurate response for other types of questions than Model B.

Model B: Model B's quality is poor for Ambiguous Questions other than this it also misinformation about GK questions like Model A.

Relevance

Model A: It responds directly to the user's request without unnecessary information.

Model B: But it does not respond directly to the user's request without unnecessary information like Model A but it also provides more and unnecessary information related to questions asked by the user.

Completeness

Model A: It mostly provides complete response but some time more explanation is needed but it provides it in short.

Model B: It covers more than users need.

Clarity

Model A: it gives a clear response which is in a simple and understanding way and well structured.

Model B: it also gives a clear response which is in a simple and understanding way and well structured but sometimes confusing for some specific topic.

Instruction Following

Model A: Follow all instructions as the user wanted to ask but sometimes it does not.

Model B: This follows all instructions and is more accurate than Model A.

Overall Performance

Based on the evaluation dataset, **[Model A/Model B] achieved the higher overall average score.**

The difference was primarily associated with Model A response on topic with accurate and completeness in a simple and organized way and cover all instructions asked by the user but Model B responded too much relevant information related to topic asked by user which is sometimes complicated to understand.

However, performance varied across categories, meaning that neither model should be considered universally superior based solely on this evaluation.

8. Common Errors

The evaluation identifies recurring response-quality issues across the dataset.

The following categories should be recorded during evaluation:

8.1 Factual Errors

Incorrect facts, inaccurate numbers, incorrect explanations, or unsupported claims.

Observed frequency: 2

8.2 Incomplete Responses

Responses that address only part of the user's request or omit important information.

Observed frequency: 2

8.3 Instruction Violations

Responses that fail to follow explicit requirements such as word count, formatting, number of examples, or requested structure.

Observed frequency: 0

8.4 Irrelevant Information

Responses that include unnecessary information or move away from the user's question.

Observed frequency: 4

8.5 Clarity Problems

Problems involving confusing explanations, poor organization, excessive repetition, or unclear wording.

Observed frequency: 2

8.6 Ambiguity Handling

Responses that make assumptions when the question does not provide enough information, rather than identifying the ambiguity or requesting clarification.

Observed frequency: 6

Error Summary

Error Type	Number of Occurrences Model A	Number of Occurrences Model B
Factual error	1	1
Incomplete answer	0	0
Instruction violation	0	0
clarity problems	2	0
Irrelevant information	0	4
Poor Ambiguity Handling	1	5

9. Key Findings

Model A is excellent for which can think like human like Ambiguous Questions response of this model is like realistic and overall this model act as human being is responding but Model B stuck here like Ambiguous Questions response it started as ai model it start using data like users personal information and more which is not good response and also for some question where one word and type of this response is enough but this model generate a long response which is time taking and unnecessary.

10. Limitations

This study has several limitations.

10.1 Limited Dataset Size

The study uses 50 prompts and 100 responses. A larger dataset would provide a broader basis for comparison.

10.2 Limited Categories

Although the dataset covers multiple categories, it does not represent every possible type of AI task.

10.3 Evaluator Subjectivity

Human evaluation involves judgment. Even with a predefined rubric, some scoring decisions may be subjective.

10.4 Single Evaluation Round

The study represents one evaluation period. AI model behavior may change over time as models are updated.

10.5 Prompt Selection

The results may be influenced by the particular prompts selected for the study.

10.6 No Universal Model Ranking

The findings should not be interpreted as proof that one AI model is universally better than another. They represent performance within this specific evaluation dataset and methodology.

11. Conclusion

This project provided practical experience in evaluating AI-generated responses using a structured human-evaluation framework.

By evaluating 100 responses across 50 prompts, the study examined multiple dimensions of AI response quality, including accuracy, relevance, completeness, clarity, instruction following, and overall usefulness.

The project also demonstrated the importance of using consistent evaluation criteria, documenting qualitative reasoning, and analyzing recurring response errors rather than relying solely on overall scores.

The completed dataset and evaluation report provide evidence of practical skills in **AI response evaluation, critical thinking, quality assessment, structured data analysis, attention to detail, and research-based judgment.**

The methodology developed through this project can also be adapted to other AI evaluation tasks, including response ranking, instruction-following assessment, content quality review, and human-feedback workflows.

Project Deliverables

- 50 evaluation prompts
- 100 AI-generated responses
- Standardized evaluation rubric
- Completed evaluation spreadsheet
- Error analysis
- Model comparison
- Final evaluation report
- Portfolio README